

Can DarkNESS see it?

A guided calculation, from the theory to the answer

DarkNESS ALP study — student handout

September 2026

The question. If dark matter decays into ALPs, and ALPs turn into X-rays in the Earth's magnetic field, would DarkNESS see any of those X-rays?

What you need. A calculator, this handout, one week. No code.

What you hand in. The one page described in Section 6, and a conversation with your advisor.

The rule for the whole task. Write your number in the box *before* you look at the answers at the back. If yours is off by more than 20%, find out why before going on. It is almost always a unit.

1 The story

Read this twice. Then explain it, out loud, to someone who is not a physicist. If you cannot, read it again.

Dark matter. Most of the matter in the universe is something we cannot see. We know it is there from how galaxies move. We do not know what it is made of. One idea: it is a heavy particle that lives for a very long time but is not quite stable.

The decay. If that heavy particle can decay, one way it could go is into two much lighter particles called ALPs (axion-like particles). Each ALP carries away half of the heavy particle's energy. If the heavy particle weighs 7 keV, each ALP has 3.5 keV. That is an X-ray energy, which is why an X-ray detector is the right tool.

Where they come from. Our galaxy sits inside a large cloud of dark matter. Decays anywhere in that cloud send ALPs toward us, and every one arrives with the same energy, 3.5 keV. So the signal is a *line*: a spike at one energy. (Decays in distant galaxies also send ALPs, but those have been stretched to lower energies on the way. We leave them out here; the line is what matters.)

The trick with the Earth. An ALP has one special property. When it crosses a magnetic field, it can turn into a photon of the same energy. The Earth is a magnet. So an ALP heading for the satellite through the Earth's field has a small chance of becoming a 3.5 keV X-ray just before it arrives. The satellite would see extra X-rays at exactly 3.5 keV, and the number would rise and fall as the orbit carries it through stronger and weaker field.

The catch. The sky is already full of X-rays from other things, and charged particles hitting the detector also make counts. The question is whether the ALP X-rays would stand out from all of that.

Write the story in your own words, five sentences. (Use the back of the page if you need to.)

2 Four ideas you need

Brightness. How many particles arrive per square centimetre, per second, from each patch of sky. A patch of sky is measured in *steradians* (sr). The whole sky is 12.6 sr. DarkNESS's 20° cone is 0.0955 sr.

Conversion probability. The chance that an ALP becomes a photon. It goes as the *square* of the ALP's coupling strength g , times the *square* of (field strength $\times$ path length through it). We call that second factor K ; it is what the orbit and the pointing control. Double the coupling: four times the chance. Double the field: four times the chance.

Counts. What the detector records.

$$\text{counts} = \text{brightness} \times \text{chance of conversion} \times \text{collecting power} \times \text{time}.$$

Collecting power is sensor area times the cone, in $\text{cm}^2 \text{sr}$. The whole chain is multiplication.

Noise. If you expect B background counts, what you actually get scatters by about $\sqrt{B}$. A signal smaller than that scatter cannot be seen. To *claim* you would have noticed a signal (at 90% confidence) it must be at least $1.28\sqrt{B}$. This one rule is the whole result.

3 The numbers you may use

Every number is in Table 1, and every one has a source. Do not change any of them. If you think one is wrong, say so, and say why.

Conversions you will need: $1 \text{ Gyr} = 3.16 \times 10^{16} \text{ s}$ and $7 \text{ keV} = 7 \times 10^{-6} \text{ GeV}$.

4 The calculation, six steps

Step 1 — The most signal nature allows

The signal is proportional to g^2 (from the conversion) times f/τ (from the decay). Multiply the two allowed maxima:

$$\Theta_{\text{max}} = g_{\text{max}}^2 \times \frac{f}{\tau}.$$

The unit is odd; that is fine. What matters is that nothing we do can make the signal bigger than this number allows.

$\Theta_{\text{max}} =$

Step 2 — How many ALPs arrive

Each heavy particle along our line of sight decays at rate f/τ . Their number per cm^2 is D/m_χ (the mass column divided by the mass of one). Each decay gives two ALPs spread over all directions, which works out to a brightness

$$I = \frac{f}{\tau} \times \frac{D}{2\pi m_\chi}.$$

Table 1: The numbers you may use.

What	Symbol	Value	Where it comes from
Strongest coupling still allowed	$g_{\max}$	$0.47 \times 10^{-10} \text{ GeV}^{-1}$	stars in globular clusters would cool too fast otherwise (Dolan et al. 2022)
Fastest decay still allowed	f/τ	$4.0 \times 10^{-3} \text{ per Gyr}$	the cosmic microwave background would look different otherwise (Planck 2018; Nygaard et al. 2021)
Heavy particle mass	m_χ	7 keV	our working choice; the line sits at $m_\chi/2 = 3.5 \text{ keV}$
Dark matter along the line of sight	D	$1.27 \times 10^{23} \text{ GeV cm}^{-2}$	the galaxy's dark-matter cloud toward its centre, averaged over the cone
Field $\times$ length, squared	K	$8.3 \times 10^3 \text{ T}^2 \text{ m}^2$	one-day average of the Dark-NESS schedule ($\sqrt{K} = 91 \text{ T m}$)
Conversion formula	P	$2.45 \times 10^{-21} (g/10^{-10})^2 K$	Yamamoto et al. 2020, eq. 2.13; we re-derived it
Sensor area	A	12 cm^2	the DarkNESS paper
Cone	Ω	0.0955 sr	20° full cone
Usable fraction of the sensor	f_{live}	0.5	half the pixels are masked
Observing time per day	t	23 400 s	6.5 h of science frames per day
Science phase		187 days	the mission plan
Sky X-ray brightness at 3.5 keV		$11.6 \times 3.5^{-1.41} \text{ per cm}^2 \text{ s sr keV}$	De Luca & Molendi 2004
Energy resolution at 3.5 keV		0.15 keV (full width)	the detector's measured resolution

Convert f/τ to *per second* and m_χ to *GeV* first.

$I =$ (per cm^2 per second per steradian)

Now say it in words, one sentence:

Step 3 — What fraction turn into X-rays

$$P = 2.45 \times 10^{-21} \times \left(\frac{g_{\max}}{10^{-10}} \right)^2 \times K.$$

$P =$

Now say it in words, one sentence: (“about one ALP in every ...”)

Step 4 — How many X-rays the detector counts

$$S_{\text{day}} = I \times P \times A \Omega \times f_{\text{live}} \times t, \quad S_{187} = 187 \times S_{\text{day}}.$$

$S_{\text{day}} =$ $S_{187} =$

Now say it in words, one sentence: (“DarkNESS would record one ALP photon about every ...”)

If this were the end we would be done. But the argument is stronger than “the signal is small”. Even a small signal can be found if the background is tiny. So:

Step 5 — How many background counts sit under the line

The line at 3.5 keV is smeared by the detector into a bump 0.15 keV wide. Take a window of $\pm$ one standard deviation around it. Its width is

$$w = 2 \times \frac{0.15}{2.355} = 0.128 \text{ keV}$$

(the 2.355 turns a full width into a standard deviation). Sky X-rays in that window per day:

$$B_{\text{sky}} = 11.6 \times 3.5^{-1.41} \times w \times A \Omega \times f_{\text{live}} \times t.$$

Charged particles add about the same again. (That is an assumption; it is what Suzaku saw.) So the total under the line is $2 \times B_{\text{sky}}$ per day, and 187 times that for the mission.

$B_{\text{sky}} \text{ per day} =$ $B_{187} =$

Step 6 — The gap

To claim a detection at 90 % confidence the signal must reach

$$S_{\text{needed}} = 1.28 \sqrt{B_{187}}.$$

Divide what is needed by what nature allows:

$$\text{gap} = \frac{S_{\text{needed}}}{S_{187}}.$$

$S_{\text{needed}} =$ $\text{gap} =$

This is the result. The signal would have to be this many times stronger than the strongest signal nature allows before DarkNESS could claim to see it. Not 2 times. Not 10 times. Write the number in words:

5 Why a bigger or longer mission does not fix it

The natural reaction is “so build a bigger one, or fly it longer”. Here is why that fails. It is the most important idea in this task.

Both the signal and the background grow with collecting power and time. Double the exposure and you get twice the signal *and* twice the background. The noise is $\sqrt{B}$, so it grows by $\sqrt{2}$, and the gap shrinks only by $\sqrt{2}$. To shrink a gap G to 1 you need G^2 times more exposure.

$G^2 =$ DarkNESS’s exposure is $1.15 \text{ cm}^2 \text{ sr} \times 4.4 \times 10^6 \text{ s}$. Times G^2 , that is:

Now put that in units a person can picture. One square metre of sensor seeing a full steradian of sky is $10^4 \text{ cm}^2 \text{ sr}$. How many years would it have to observe?

years =

What about being cleverer instead of bigger? Take every unfinished idea at its most generous:

Idea	Best it could do to the gap
Point where the field is strongest, all the time	$\div 13$ (the field has a hard maximum, about 330 Tm along any line from 420 km)
Observe in sunlight too, doubling the time	$\div 1.4$
Somehow have no particle background at all	$\div 1.4$
Use a looser bound on the decay rate (another survey allows $5\times$ more)	$\div 5$
Replace the CCD with a detector $30\times$ sharper in energy	$\div 5.5$

All of them together: $\div 1000$ at the very most, leaving a gap of a few hundred. Still no photon.

So the conclusion is not about DarkNESS. It is about the idea. Any X-ray telescope in Earth orbit, using the Earth as its magnet, is too far from the allowed signal by a factor no engineering closes. That is a real result, and you found it with one page of arithmetic.

6 What to hand in

One page, in this order:

1. The story of Section 1 in your own words, five sentences.
2. The six numbers from Section 4, each with its formula and its units.

3. This sentence, completed with your numbers: “*At the largest signal current limits allow, DarkNESS would record _____ photons in its mission against _____ background counts under the line; the signal would need to be _____ times larger to be seen, and more exposure closes that gap only as its square root.*”
4. One thing that surprised you.

Then talk it through with your advisor. Expect these questions:

- Which step is the bottleneck? (Not the number of ALPs.)
- Why does more observing time not help?
- What *would* change the answer?

7 Traps

- **Units.** Gyr to seconds; keV to GeV. If Step 2 is off by 10^{16} or 10^6 , this is why.
- **g versus g^2 .** The conversion uses the square. Forgetting it makes the signal 2×10^{10} too large.
- **Steradians.** Brightness is per steradian. Multiply by the cone, not the whole sky.
- **“More ALPs means more signal.”** No. Step 2 says there are hundreds of millions per cm^2 per second. Step 3 is the wall.
- **“A bigger telescope would see it.”** Only as the square root, and the background grows with it.
- **Making the answer come out.** If you get a gap of 10, you have made a mistake, not a discovery. Find it.

Check your numbers

Look here only after you have written yours in the box.

Step	Answer	In words
1	$8.8 \times 10^{-24} \text{ GeV}^{-2} \text{ Gyr}^{-1}$	the ceiling; nothing can exceed it
2	$3.7 \times 10^8 \text{ per cm}^2 \text{ s sr}$	370 million ALPs cross every square centimetre every second. The signal is not small because ALPs are few.
3	4.5×10^{-18}	about one ALP in 2×10^{17} becomes a photon. This is the bottleneck: a weak magnet and a tiny coupling.
4	2.2×10^{-5} per day; 4.1×10^{-3} in 187 days	one ALP photon about every 125 years. Not one in the mission.
5	3.4×10^3 per day (sky); 1.25×10^6 in 187 days (sky + particles)	
6	$S_{\text{needed}} \approx 1400$; gap $\approx 3.5 \times 10^5$	the signal would have to be a third of a million times stronger
§5	$G^2 = 1.2 \times 10^{11}$; $6 \times 10^{17} \text{ cm}^2 \text{ sr s}$; about 20 million years	one square metre steradian for twenty million years

The full analysis, which fits the whole spectrum and all the backgrounds at once, gives a gap of 3.8×10^5 . Your page of arithmetic agrees with it to 10%.